

Notes: Ellison & Fudenberg (QJE, 1995)

Word-of-Mouth Communication and Social Learning

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1 Big picture

- **Question.** How does word-of-mouth communication aggregate information, and when does it produce conformity, diversity, efficient learning, or inefficient herding?
- **Setting.** A continuum of agents repeatedly chooses between two products, technologies, or practices, f and g .
- **Behavior.** Agents are boundedly rational: only a fraction reevaluates each period, they sample N others, and they use simple average-payoff comparisons.
- **Main equal-payoff result.** When choices are equally good, small samples tend to generate conformity, while larger samples generate diversity.
- **Main unequal-payoff result.** When one product is better more often, inefficient herding, efficient social learning, and diversity are all possible.
- **Takeaway.** Less communication can be more efficient because limited samples create enough lock-in to eliminate inferior choices, while overly broad samples can sustain persistent diversity.

2 Incremental contribution of the paper

- **From rational herding to boundedly rational word-of-mouth.** The paper models casual communication rather than fully Bayesian sequential learning.
- **Communication structure as the key parameter.** The sample size N determines whether society converges, diversifies, or herds incorrectly.
- **Counterintuitive efficiency result.** More information is not always better; limited communication may deliver efficient social learning.
- **Must-see restriction.** Agents cannot switch to alternatives they never hear about, creating endogenous popularity weighting.
- **Local-dynamics method.** A Markov endpoint lemma turns a complex stochastic system into threshold conditions.

3 Model objects and measurement

3.1 Population, payoffs, and shocks

Agents choose between f and g at dates $t = 1, 2, \dots$. The state x_t is the fraction using g .

$$\text{payoff to } f = f_t + \varepsilon_{ift}, \quad \text{payoff to } g = g_t + \varepsilon_{igt}, \quad \theta_t = g_t - f_t.$$

The common shock is $\theta_t = \theta > 0$ with probability p and $-\theta$ with probability $1 - p$. Idiosyncratic shocks have standard deviation σ .

Interpretation: Idiosyncratic noise makes sampling useful; common shocks make broad samples potentially misleading.

3.2 Inertia and sampling

A fraction α reevaluates each period. Reevaluating agents sample N others and choose the product with the higher average payoff among observed experiences. A player cannot switch to an alternative absent from the sample.

Interpretation: N is the communication technology; smaller N implies more local and popularity-sensitive learning.

4 Models and theoretical specifications

4.1 Local endpoint approximation

Near $x = 0$,

$$\begin{aligned} x_{t+1} &= x_t[1 - \alpha + \alpha(N+1)q^+(N)] + o(x_t) \quad \text{with prob. } p, \\ x_{t+1} &= x_t[1 - \alpha + \alpha(N+1)q^-(N)] + o(x_t) \quad \text{with prob. } 1 - p. \end{aligned}$$

The key threshold statistic is

$$Q(\alpha, N) = \frac{-\log\{1 + \alpha[(N+1)q^-(N) - 1]\}}{\log\{1 + \alpha[(N+1)q^+(N) - 1]\}}.$$

Interpretation: Q compares forces pushing the state away from or toward an endpoint.

4.2 Equal-payoff and unequal-payoff regimes

When $p = 1/2$, $Q(\alpha, N) > 1$ implies conformity and $Q(\alpha, N) < 1$ implies diversity. When $p > 1/2$, intermediate values of $Q(\alpha, N)$ imply efficient social learning; too-low or too-high values imply diversity or inefficient herding.

5 Theoretical design and credibility

The model isolates information flow: there are no direct payoff externalities from choosing what others choose. The continuum population makes aggregate dynamics deterministic conditional on common shocks. The main limitation is that sampling and behavior are exogenous and not empirically estimated.

6 Figures and extracted theorem blocks

6.1 Lemma 1: Local Markov condition for lock-in

Read: Lemma 1 gives the local condition for whether a Markov process can converge to zero.

Economic message: It provides the technical foundation for classifying endpoint convergence.

the system. Given the complexity of our model, the ability to work with a linear approximation will be an advantage.

LEMMA 1. Let x_t be a Markov process on $(0,1)$ with

$$x_{t+1} = \begin{cases} H_1(x_t) & \text{with probability } p \\ H_2(x_t) & \text{with probability } (1-p) \end{cases}.$$

Suppose that in a neighborhood of 0, $H_i(x_t) = \gamma_i x_t + o(x_t)$, with $\gamma_2 < 1 < \gamma_1$.

(a) If

$$\frac{p}{1-p} > -\frac{\log(\gamma_2)}{\log(\gamma_1)},$$

then x_t cannot converge to 0 with positive probability.

(b) If

$$\frac{p}{1-p} < -\frac{\log(\gamma_2)}{\log(\gamma_1)},$$

then there are strictly positive δ and ϵ such that $\text{Prob}[x_t \rightarrow 0 | x_0 \leq \delta] \geq \epsilon$.

(c) If

$$\frac{p}{1-p} > -\frac{\log(\gamma_2)}{\log(\gamma_1)},$$

there is an $x^* > 0$ such that for all $x_0 > 0$, $\text{Prob}[x_t < x^* \forall t | x_0] = 0$.

Lemma 1: local Markov condition

6.2 Figure I and Figure II: Threshold sample sizes under equal payoffs

Read: Figure I plots $N^*(\alpha)$ for $\eta = 2$. Figure II shows how the threshold changes with common-shock strength η .

Economic message: Small samples produce conformity; common shocks make conformity more likely.

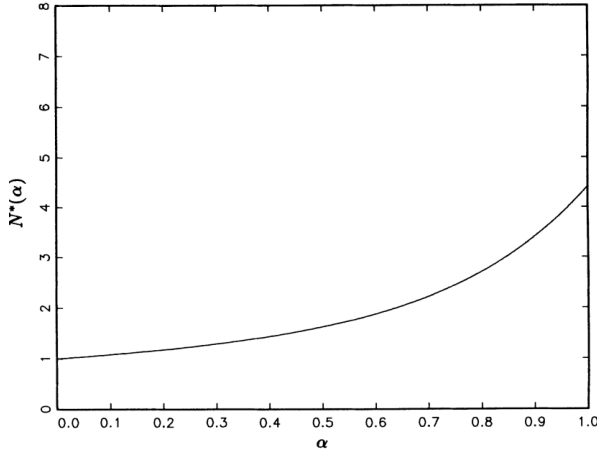


FIGURE I

(a) Figure I: threshold $N^*(\alpha)$

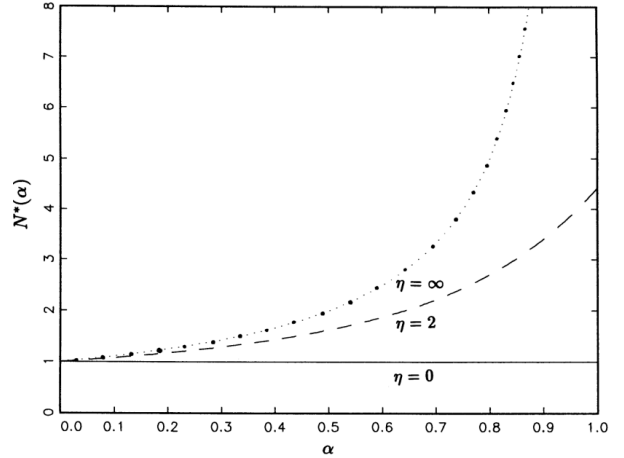


FIGURE II

(b) Figure II: role of η

6.3 Theorem 1: Conformity versus diversity when $p = 1/2$

Read: Theorem 1 proves the threshold result for equally efficient choices.

Economic message: Herding can arise without payoff superiority or strategic complementarities.

common shock obtained by g -users will cause a large number of players to switch to g , while unfavorable draws will cause only a few players to switch away from g . As a result, unpopular technologies will not die out, and the system will exhibit diversity.

THEOREM 1. When $p = 1/2$,

(a) The state cannot converge to an endpoint if

$$1 > \frac{-\log(1 + \alpha((N+1)q^-(N) - 1))}{\log(1 + \alpha((N+1)q^+(N) - 1))} \equiv Q(\alpha, N).$$

(b) The state converges to an endpoint with probability 1 if $1 < Q(\alpha, N)$. In this case the system has positive probability of converging to either endpoint.

(c) There is a unique $N^*(\alpha)$ such that $Q(\alpha, N^*(\alpha)) = 1$. Further, N^* is increasing in α , and $N^*(\alpha) > 1 \forall \alpha$ provided that $\theta > 0$.

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Theorem 1

6.4 Figure III and Figure IV: Learning regions with unequal payoffs

Read: Figure III shows diversity, efficient learning, and inefficient herding regions. Figure IV shows how higher p expands efficient learning.

Economic message: The superior payoff creates drift toward the better technology, but communication structure determines whether the drift is enough.

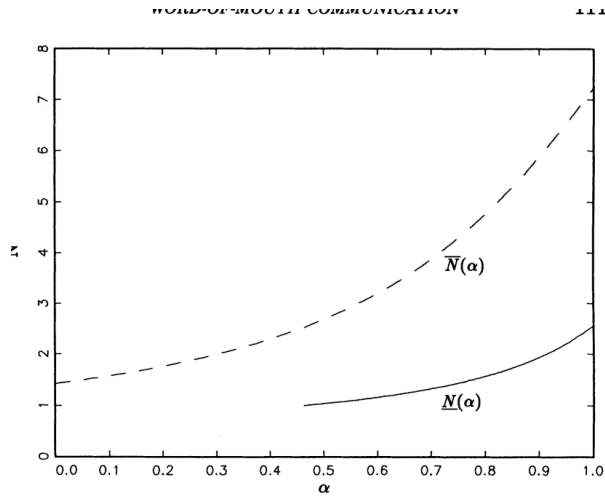
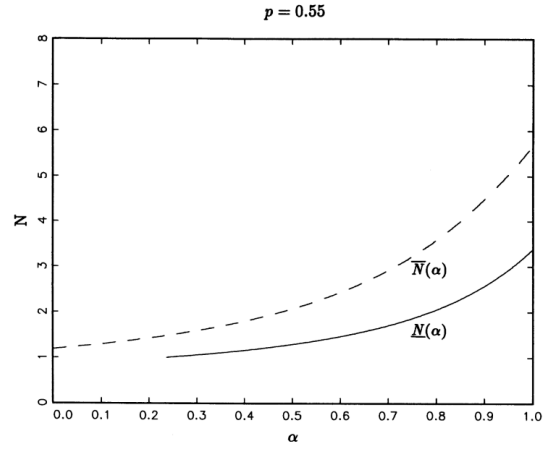


FIGURE III

(a) Figure III: three regimes

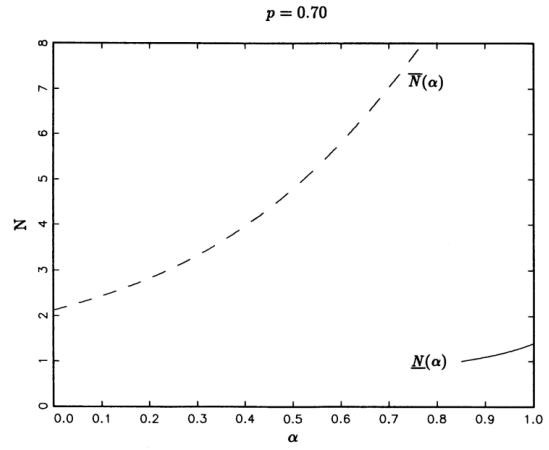


FIGURE IV

(b) Figure IV: effect of p

6.5 Theorem 2: Inefficient herding, efficient social learning, and diversity

Read: Theorem 2 classifies the three regimes when $p > 1/2$.

Economic message: This is the central result: limited word-of-mouth can produce efficient social learning.

the word-of-mouth communication is extensive, and efficient social learning when players receive less information.

THEOREM 2. Suppose that $p > 1/2$.

- (i) If $\lfloor p/(1-p) \rfloor < Q(\alpha, N)$, the system exhibits inefficient herding: the state converges to an endpoint with probability one, and converges to everyone using the inferior product f with positive probability from all initial conditions.

8. See, e.g., Griliches [1957, 1994] and Mansfield [1968] for empirical studies.

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- (ii) If $\lfloor (1-p)/p \rfloor < Q(\alpha, N) < \lfloor p/(1-p) \rfloor$, the system exhibits efficient social learning; i.e., it converges with probability one to the state where everyone uses g .
- (iii) If $\lfloor (1-p)/p \rfloor > Q(\alpha, N)$, the system exhibits diversity, as the state does not converge to either endpoint. Moreover, if $\eta \neq 0$, the system does not converge to any deterministic limit; more precisely, $\text{Prob}(\exists x \text{ s.t. } x_t \rightarrow x) = 0$.

Theorem 2

6.6 Theorem 3 and Figure V: Popularity weighting

Read: Theorem 3 extends the baseline to explicit popularity weighting; Figure V shows how weighting changes learning regions.

Economic message: Popularity weighting can make larger samples efficient by correcting overreaction to rare alternatives.

and let $N^*(\alpha, m)$ be the solution to $Q(\alpha, N, m) = 1$. (Calculations similar to those of Section IV show that a solution exists and is unique.) As in Section V, $N = N^*(\alpha, m)$ is the one sampling rule that guarantees efficient social learning uniformly over all values of p .

THEOREM 3. Suppose that $p > 1/2$.

- (i) If $[p/(1-p)] < Q(\alpha, N, m)$, the state converges to an endpoint with probability one, and exhibits inefficient herding, as the state converges to the inferior choice with positive probability from initial conditions sufficiently close to 0. If m is large enough, this inefficient herding can occur with probability one for some initial conditions.
- (ii) If $[(1-p)/p] < Q(\alpha, N, m) < [p/(1-p)]$, there is efficient social learning: the system converges with probability one to the state where everyone uses g .
- (iii) If $[(1-p)/p] > Q(\alpha, N, m)$, the system does not converge to either endpoint. Moreover, if $\eta \neq 0$, the system does not converge to any deterministic limit; more precisely, $\text{Prob}(\exists x \text{ s.t. } x_t \rightarrow x) = 0$.
- (iv) $Q(\alpha, N, m)$ is increasing in m wherever it is defined and nonnegative, so that a small increase in m , holding other parameters constant, either causes a change from case (iii) to case (ii) or a change from case (ii) to case (i), or has no effect on which of the three cases prevails. Moreover, the uniformly optimal sampling rule $N^*(\alpha, m)$ is increasing in m .

(a) Theorem 3

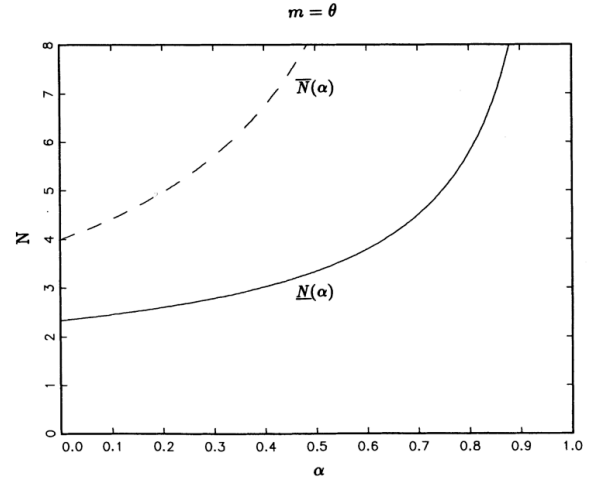
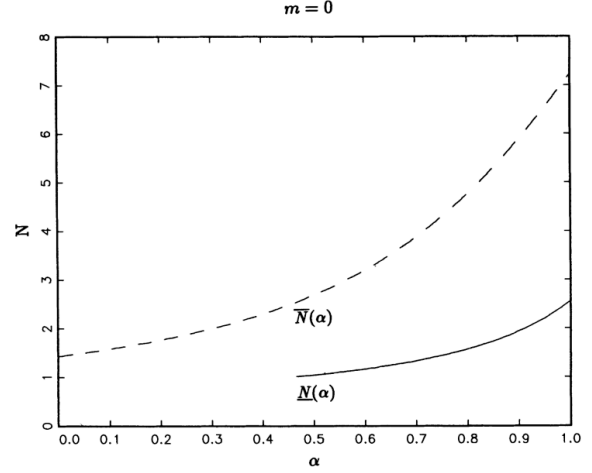


FIGURE V

(b) Figure V

6.7 Social memory equation block

Read: Social memory lets reports aggregate experiences from multiple past periods, reducing common-shock noise.

Economic message: Memory can make broader communication nearly efficient.

$$(3) \quad x_{t+1} = (1 - \alpha)x_t + \alpha \text{Prob}(\hat{g} > \hat{f} | N, x_t),$$

where the probability term in (3) is the probability that the average realized payoff to g exceeds the average realized payoff to f in a sample of $N + 1$ players drawn from a population where x_t of the players use g . (If no players in the sample use g , g 's average realized payoff in the sample is minus infinity.)

In a neighborhood of $x = 0$, equation (3) can be approximated by

$$x_{t+1} = x_t[1 - \alpha + \alpha(N + 1)(p\bar{q}^+(N) + (1 - p)\bar{q}^-(N))] + o(x_t),$$

where $\bar{q}^+(N)$ is the probability that a single draw from the favorable distribution of g exceeds the average of N draws from the distribution of f .

Thus, the state converges to 0 if $p\bar{q}^+(N) + (1 - p)\bar{q}^-(N) < 1/(1 + N)$, and the state cannot converge to zero if the reverse inequality holds. If $p < 1/2$, this inequality is satisfied for $N = 1$. To see this, note that

$$\begin{aligned} p\bar{q}^+(1) + (1 - p)\bar{q}^-(1) &= p^2\Phi\left(\frac{\theta}{\sigma\sqrt{2}}\right) + 2p(1 - p)\frac{1}{2} \\ &\quad + (1 - p)^2\left(1 - \Phi\left(\frac{\theta}{\sigma\sqrt{2}}\right)\right). \end{aligned}$$

Social memory condition

7 Short summary

- Word-of-mouth communication transmits information through random samples of other agents' experiences.
- Small samples make rare products hard to hear about and promote conformity.
- Large samples make rare products more likely to be observed and promote diversity.
- When one technology is superior, intermediate sampling generates efficient social learning.
- The incremental contribution is to show that communication breadth directly affects social efficiency under bounded rationality.

8 Conclusion

8.1 Core conclusion

Word-of-mouth communication can generate conformity, diversity, efficient learning, or inefficient herding depending on communication structure.

8.2 Mechanism

The must-see restriction makes popularity matter. Limited sampling can drive unpopular technologies out of the market, while broad sampling can prevent extinction and sustain diversity.

8.3 Incremental contribution

The paper's central contribution is to show that the architecture of informal communication is itself an economic force. More communication can reduce efficiency when common shocks are important.

8.4 Limitations and extensions

The model is theoretical, assumes exogenous sampling, and uses simple behavioral rules. Empirical network data, endogenous search, strategic communication, persistent heterogeneity, and richer memory are natural extensions.

8.5 Takeaway

Social learning is shaped not only by what people know, but by how widely and how locally they communicate.